

STANDARD U-100 INSULIN CALCULATOR HELP

The CHLA Insulin Calculator will accurately determine the correct dose of rapid-acting insulin to be administered based on a specific patient order, the grams of carbohydrates eaten, and the measured blood glucose.

Insulin/Carbohydrate Ratio (ICR):

- 1 unit of insulin covers **X** grams of carbohydrates eaten
- Allowed range: 1 – 100 gm:1 unit
 - “Usual” ICR (~10th – 90th %ile) varies by age
 - 0 - 5y: 10 – 50
 - 6 - 11y: 5 – 30
 - ≥ 12y: 4 – 12



ordered value; click to confirm → ✓



indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 8 | HIGH: > 80
- 6 - 11y: LOW: < 4 | HIGH: > 75
- ≥ 12y: LOW: < 2 | HIGH: > 35

Correction Factor (CF) -- also called Insulin Sensitivity Factor (ISF):

- 1 unit of insulin reduces a Blood Glucose elevated over Target Blood Glucose by **X** mg/dL
- Allowed range: 4 – 300 mg/dL:1 unit
 - “Usual” CF/ISF (~10th – 90th %ile) varies by age
 - 0 - 5y: 50 – 200
 - 6 - 11y: 25 – 100
 - ≥ 12y: 15 – 50



ordered value; click to confirm → ✓



indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 40 | HIGH: > 200
- 6 - 11y: LOW: < 15 | HIGH: > 200
- ≥ 12y: LOW: < 8 | HIGH: > 120

Target Blood Glucose:

- blood glucose target above which correction must be applied
- Age, individual and context-dependent
- Allowed range: 100 – 250 mg/dL

- “Usual” Target BG (~10th – 90th %ile):

- 0 - 5y: 150 – 200
- 6 - 11y: 130 – 180
- ≥ 12y: 120 – 150



ordered value; click to confirm →



indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 130 | HIGH: > 250
- 6 - 11y: LOW: < 120 | HIGH: > 200
- ≥ 12y: LOW: < 120 | HIGH: > 200

****The calculator is not enabled for input of carbs eaten or measured blood glucose until all above parameters are confirmed by clicking the or***

Rapid-acting insulin in past 2 hrs?:

- *If yes, will not include any correction dose for high blood glucose in total dose calculation in order to prevent insulin stacking for insulin on board*
- Calculated dose will include carb coverage only
 - Defaults to YES if MAR shows rapid-acting insulin charted within past 2 hrs
 - Defaults to NO if recent dose not detected
 - No recent dose
 - Dose given but not charted (home, pump, other)
- response may be changed based on circumstance

Carbohydrates Eaten:

- *Enter the grams of carbohydrates eaten*
- calculator will accept any value from 0 – 300 gm
 - carbs > 250 gm; verify and correct if data entry was incorrect
 - Carb coverage will only be provide up to 250 gms
 - unacceptable entry which must be corrected

Blood Glucose for Calculation:

- *Enter the current glucose measurement*
- calculator will accept any value from 1 – 2,000 mg/dL
 - BG > 600 mg/dL; verify and correct if data entry was incorrect
 - Correction is only performed for BG up to 600
 - BG > 70 mg/dL: hypoglycemia

- STOP – Calculator will not calculate dose
- DO NOT GIVE INSULIN
- requires rescue intervention for low BG

✗ unacceptable entry which must be corrected

To complete calculation of dose to administer:

- *Assure all parameters and patient values are valid*
- Click button
- Dose will be displayed, **rounded** to half or whole units per order increments, here:

Calculated Dose to Administer


 A green arrow points from the left towards a red-outlined rectangular input field. To the right of the input field is the text "units".

- Transfer calculated dose to MAR after independent 2nd witness check and dose has been administered
- **REMEMBER WHEN CHARTING**
 - *Change time from default to time actually administered*
 - *Change dose from “1 dose” to “X units” actually administered*